Morphological Image Processing & Segmentation

Image Processing â¢ Unit 3: Morphological Processing â¢ Study Notes

Erosion, dilation, opening, closing, hit-or-miss transform, and boundary detection.

1. Basic Morphological Operations

Structuring elements: shape, size, and origin definition.

Erosion: shrinking binary image objects according to structuring element.

Dilation: expanding boundaries and bridging small spatial gaps.

Opening (erosion followed by dilation) and Closing (dilation followed by erosion).

2. Morphological Algorithms

Boundary extraction: difference between original set and eroded set.

Hole filling: conditional dilation based on complement mask.

Connected components labeling for region segmentation.